ANALYZE BEHAVIOR AND EMOTIONS IN TELEGRAM AND WHATSAPP MESSAGES TO DETECT FAKE USERS.

User emotions in Telegram and WhatsApp Messages to detect fake Users.

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Project Proposal Report

B.Sc. (Hons) in Information Technology Specializing in Cyber Security

Sri Lanka Institute of Information Technology Sri Lanka

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously publish or written by another person expect where the acknowledgement is made in the text.

Name	Student ID	Signature
P.S.N.A Rupasinghe	IT22560230	

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

.....

Signature of the Supervisor:
(Mr. Deemantha Siriwardana)

Date: 27/08/2025

.....

Signature of the Co-supervisor:
(Miss. Ayesha Wijesooriya)

Date: 27/08/2025

Abstract

This research proposes a machine learning-driven mobile application to safeguard children from scammers and fake accounts on WhatsApp and Telegram, addressing a critical gap in online safety for vulnerable users in Sri Lanka. The rising use of these encrypted platforms for online learning has heightened exposure to manipulative behaviors, yet existing studies focus on content-level threats, neglecting behavioral and emotional cues. The application captures messages, converts Singlish to English using a fine-tuned Large Language Model (LLM) like BERT, and analyzes emotional tones and behavioral patterns (e.g., message frequency, timing) with ML techniques such as Random Forest and LSTM. It generates real-time threat alerts low, medium, or high risk via an interactive parent dashboard.Projected outcomes include an F1-score exceeding 0.85 in detecting fake accounts, validated with synthetic and public datasets. This novel, region-specific solution enhances child safety, with commercialization potential targeting schools and safety organizations, supported by a 25,500 LKR budget. The study advances multi-modal threat detection, aligning with Sri Lanka's cybersecurity goals.

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List of Abbreviations

Abbreviation		Description
API	-	Application Programming Interface
BERT	-	Bidirectional Encoder Representations from Transformers
LLM	-	Large Language Model
LSTM	-	Long Short-Term Memory
NLP	-	Natural Language Processing
ONNX	-	Open Neural Network Exchange

I. INTRODUCTION

III.1 Background and Literature Review

Messaging platforms such as WhatsApp and Telegram have become essential for daily communication, particularly among young users. While these platforms enable instant and interactive conversations, they are also exploited by malicious actors who create fake or harmful accounts to emotionally manipulate children, spread harmful content, or encourage unsafe actions. Detecting such risks is challenging because traditional rule-based or keyword-matching approaches cannot capture the subtle variations in emotional tones expressed in human conversations.

Recent studies on emotion detection in text highlight that machine learning and deep learning models provide significant improvements over older techniques. Shukla and Dwivedi [1] compared machine learning (ML), deep learning (DL), and hybrid methods for text-based emotion recognition and found that no single approach provides a complete solution, though models like BERT achieve strong accuracy across diverse datasets. Similarly, Koufakou et al. [2] evaluated multiple datasets including COVID-19 survey data, WASSA-21 essays, and GoEmotions Reddit comments and demonstrated that transformer-based models such as RoBERTa consistently outperform traditional ML models in capturing nuanced emotions like anger, sadness, and fear. Building on this, Kamath et al. [3] proposed EmoRoBERTa, a fine-tuned RoBERTa model trained on the GoEmotions dataset, which improved upon baseline BERT results by recognizing 27 distinct emotions with higher precision.

Despite these advancements, most prior work focuses on flat emotion categories (anger, sadness, fear, joy) and does not capture their intensity levels. For instance, anger expressed mildly is very different from explosive rage, but current systems often treat both as the same class. This lack of level-based emotion classification limits the ability to detect escalating threats or manipulative intent in conversations. Moreover, research has largely targeted open platforms like Twitter or Reddit, with limited exploration of encrypted and closed-group platforms such as WhatsApp and Telegram, which are widely used by children in Sri Lanka for both education and social interaction

Therefore, this research addresses the gap by developing a mobile application that applies level-based emotional analysis (low, medium, high intensity) to WhatsApp and Telegram conversations. By detecting not only the presence of emotions but also their severity, the system aims to flag high-risk interactions, generate real-time parental alerts, and empower guardians to block harmful contacts. This approach builds on recent advances in transformer models while tailoring the solution to the unique challenges of private, encrypted messaging environments.

III.2 Research Gap

While numerous studies have advanced text-based emotion detection using machine learning and deep learning models, most focus only on identifying basic emotions such as anger, sadness, joy, or fear. These approaches treat emotions as single, flat categories and overlook the varying intensity levels within an emotion. For instance, anger can manifest as mild irritation, standard anger, or explosive rage, yet existing models rarely distinguish between these levels. This limits the usefulness of emotion detection in real-world conversations, where the strength of the emotion often signals the seriousness of manipulation or risk. In addition, prior work is largely centered on datasets from general social media platforms such as Reddit or Twitter, with little emphasis on messaging platforms like WhatsApp and Telegram, which are widely used by children and young users.

Furthermore, most prior research demonstrates strong performance in emotion recognition but does not extend this to level-based classification. While current models can say "this message is angry," they do not indicate whether it is a low, medium, or high level of anger. This distinction is critical for child protection, because higher emotional intensity often aligns with greater psychological pressure or manipulation risk. Given the increasing use of WhatsApp and Telegram among vulnerable groups, the lack of level-based emotional analysis tailored for these platforms represents a significant research gap. Addressing this gap will allow more precise detection of harmful conversations and provide better safeguards for children in real-time communication.

Table I-1 - Research Gap in tabular view

Research Paper	Emotion Analysis	Level-Based Emotional Analysis	Platform Specific (WhatsApp/Telegram)
Research [1]	Implemented	Not Implemented	Not Implemented
Research [2]	Implemented	Not Implemented	Not Implemented
Research [3]	Implemented	Not Implemented	Not Implemented
Proposed Solution	Implemented	Implemented	Implemented

III.3 Research Problem

During the COVID-19 pandemic, the increased use of messaging applications such as WhatsApp and Telegram among school children in Sri Lanka under the age of 16 created both opportunities and risks. While these platforms supported online learning and peer communication, they also became a gateway for scammers and malicious actors to exploit vulnerable users.

Children are particularly at risk because they lack the maturity and awareness to identify suspicious behavior, fake accounts, or harmful conversations. Scammers often send deceptive messages, fake links, or emotionally manipulative texts that can mislead children into:

- Clicking on phishing links that may steal personal information.

- Downloading malicious files disguised as study materials or entertainment.
- Engaging in harmful chats where they can be emotionally manipulated or coerced.
- Being exposed to inappropriate or sensitive content can negatively affect their psychological well-being.

Unlike adults, children may not fully understand the long-term consequences of interacting with suspicious accounts. For example, they may trust a fake identity posing as a classmate or teacher, leading them to share sensitive personal details or follow harmful advice. These manipulative message patterns and emotional cues can result in emotional harm, behavioral changes, and even long-term safety risks.
At present, parents and teachers cannot continuously monitor children's private chat activities due to both practical limitations and privacy concerns. Therefore, there is a pressing need for a smart and user-friendly solution that can automatically detect suspicious behavior and emotional manipulation in text messages. Such a system should alert users or guardians when harmful or deceptive patterns are detected, without invading the child's privacy by storing or exposing entire conversations.

Thus, the core research problem is:
How can emotional and behavioral analysis of text messages be used to detect fake, suspicious, or harmful accounts in messaging platforms like WhatsApp and Telegram, in order to protect children from online scams and emotional manipulation?

II. OBJECTIVES

III.1 Main Objective

To develop a cross-platform mobile application that integrates advanced multi-modal machine learning algorithms, including natural language processing (NLP), computer vision, and level-based emotional analysis, alongside sandbox-based dynamic analysis techniques, to proactively identify and mitigate a range of cybersecurity threats within popular messaging platforms such as WhatsApp and Telegram. This encompasses the detection of spam and malicious links through simulated URL behavior analysis, identification of document-based malware (e.g., in PDFs, DOCs, and APKs) via runtime execution monitoring, detection of emotional tones at different intensity levels (e.g., low, medium, high anger, sadness, fear, etc.) to flag fake or harmful accounts, and filtering of sensitive multimedia content (e.g., images and videos containing nudity, violence, or explicit material) using real-time classification models. The overarching goal is to enhance digital safety for vulnerable users, particularly minors in educational and social contexts, by providing intuitive alerts, user-friendly interfaces, and privacy-preserving mechanisms that prevent platform misuse, reduce exposure to online risks, and promote secure communication practices in alignment with national cybersecurity standards in Sri Lanka.

III.2 Sub-Objectives

To investigate and model level-based emotional tones in WhatsApp and Telegram messages by

leveraging machine learning techniques, including natural language processing for sentiment and emotion analysis, with a focus on distinguishing emotions at different intensity levels (e.g., low, medium, and high anger, sadness, or fear). The goal is to detect and classify fake or suspicious accounts through advanced emotion classification models, evaluate their performance using accuracy and other relevant metrics, and provide visual interpretations through interactive dashboards, ultimately contributing to enhanced threat detection and safer user interactions within the overarching mobile application framework.

III. METHODOLOGY

The methodology develops a mobile application module to protect children from scammers on WhatsApp and Telegram by analyzing messages for fake account detection. Using Python and libraries like Hugging Face Transformers and Scikit-learn, the process includes four stages: Message Capture, Language Processing, ML Analysis, and Alert Generation.

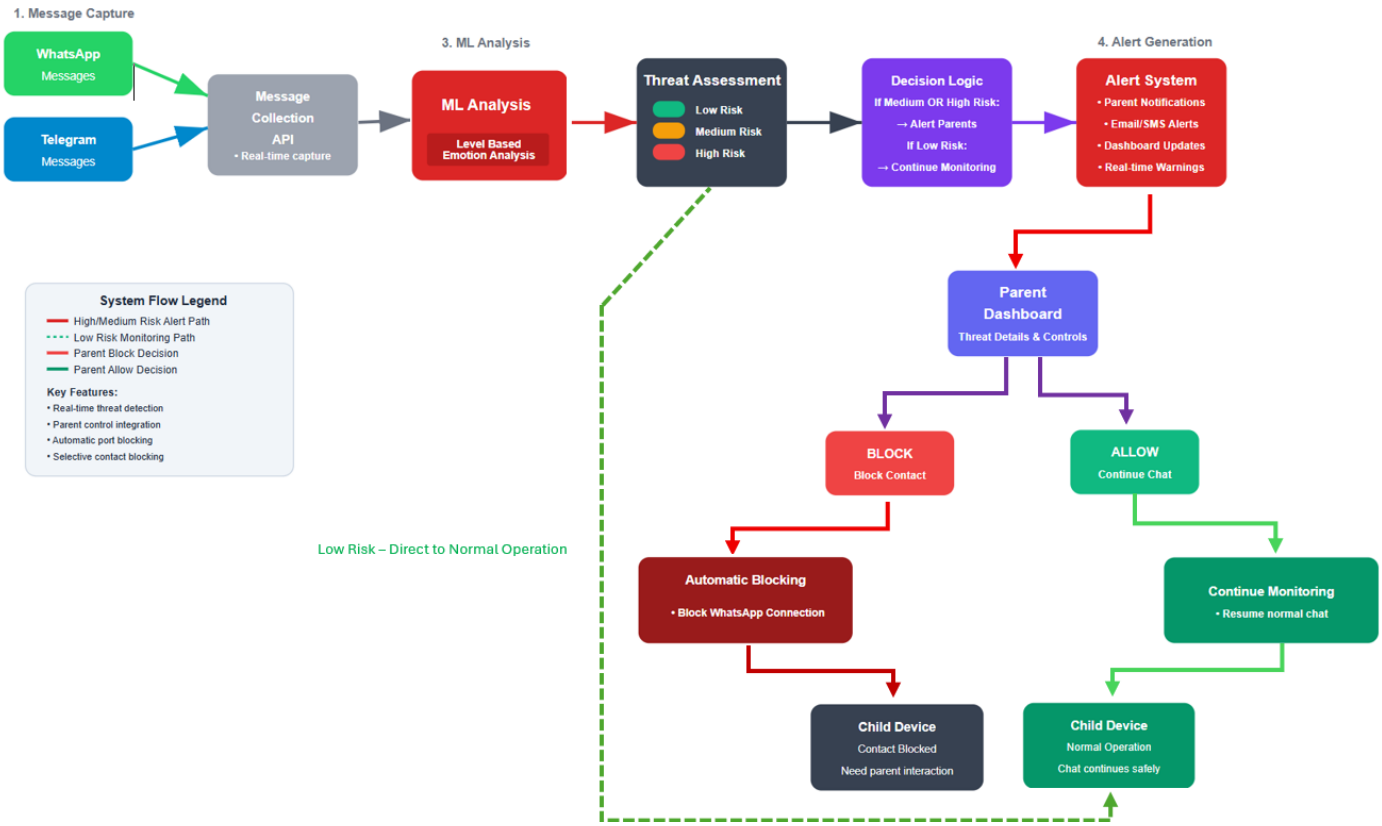

Figure 1 - System Diagram

Message Capture

Messages are collected in real-time from the child's WhatsApp and Telegram accounts using APIs (e.g., WhatsApp Business API, Telegram Bot API) with parental consent, extracting text and metadata from simulated or anonymized datasets for testing.

ML Analysis

English messages are analyzed for emotional tones and their intensity levels (e.g., mild, medium, or high anger, sadness, joy, fear, etc.) using transformer-based models such as BERT or DeBERTa for sentiment and level-based emotion classification. Trained on public and synthetic datasets, the model outputs threat levels (low, medium, high) and is deployed with TensorFlow Lite for real-time, on-device processing.

Alert Generation

The dashboard visualizes level-based emotional insights (e.g., high anger, fear, or sadness) in a simple, interpretable format. In addition, if a parent chooses to block the suspicious contact, the system automatically restricts WhatsApp communication by blocking the relevant ports/Network, preventing further messages from that account on the child's device.

This approach ensures a scalable, privacy-compliant solution that combines real-time emotional risk detection with parental control. The design will be validated through testing and stakeholder feedback to safeguard children.

III.1 Requirement Gathering

The requirement gathering process for the development of a machine learning-driven module to analyze level-based emotional tones in WhatsApp and Telegram messages for detecting fake or suspicious accounts involved a comprehensive approach to ensure alignment with the project's objectives. This process encompassed the following key activities:

Extensive Literature Review: A thorough review of recent academic papers, journal articles, and technical reports on natural language processing (NLP), sentiment analysis, and emotion detection in messaging platforms was conducted. This review focused on understanding current methodologies for detecting fake accounts, limitations of existing systems, and emerging trends in chat-based threat detection, particularly for WhatsApp and Telegram. Insights from this analysis helped identify gaps in real-time, level-based emotion analysis, guiding the design of a novel, platform-specific solution.

Stakeholder Consultation: Engagements with cybersecurity experts, child safety advocates, and mobile application developers were carried out through interviews, surveys, and workshops. These consultations provided practical perspectives on the challenges of identifying deceptive accounts in messaging apps, user expectations for privacy, and the need for lightweight, real-time detection systems suitable for mobile devices. Feedback from stakeholders, including insights from Sri Lanka CERT|CC and child protection agencies, ensured the module addresses real-world safety concerns for vulnerable users, such as minors.

System Requirements Analysis: Detailed technical specifications were documented, focusing on the module's requirements for NLP-based sentiment and emotion analysis, fine-grained classification of emotions into intensity levels (e.g., low, medium, high), and machine learning model performance. This analysis also covered scalability for real-time processing on mobile devices, privacy-preserving techniques for encrypted platforms, and integration with the broader application's user interface to deliver intuitive alerts and dashboards.

Use Case Scenarios: Multiple use case scenarios were developed to model how the module will function in diverse WhatsApp and Telegram environments. These scenarios included identifying fake accounts spreading spam, detecting manipulative language in group chats, and flagging emotionally harmful interactions at different intensity levels. By simulating real-world communication patterns, these use cases ensured the module addresses practical challenges and delivers effective, user-centric threat detection in dynamic messaging contexts.

III.2.2-i Literature Review of Key Studies:

The following key studies provide foundational insights into the use of natural language processing (NLP), sentiment analysis, and behavioral analytics for detecting suspicious or fake accounts in social media and messaging platforms, with a focus on their relevance to the proposed module for WhatsApp and Telegram.

Research [4]: Mansoori et al. (2020)
This study explores sentiment analysis using NLP to detect suspicious activities on Twitter and Facebook, employing a Long Short-Term Memory (LSTM) network to analyze emotional tones in social media posts. The authors emphasize the effectiveness of LSTM in capturing long-term dependencies in text, achieving high accuracy in identifying malicious behavioral profiles. However, the study focuses on public social media platforms and does not address the unique challenges of encrypted messaging apps like WhatsApp and Telegram, such as limited data access and real-time processing constraints. Additionally, it lacks integration of behavioral features like message timing or frequency, which are critical for identifying fake accounts in chat-based environments

Research [5]: Gagandeep and Verma (2024)
This work proposes an anomaly detection system for social media posts using NLP and behavioral profiling to flag suspicious activities. The methodology combines sentiment analysis with heuristics to detect malicious behavior, achieving improved precision and recall compared to traditional approaches. While the study highlights the importance of behavioral features, it primarily focuses on open platforms like X and does not address the specific context of private messaging apps. The lack of focus on real-time, mobile-optimized detection and the absence of emotional tone analysis tailored to encrypted platforms limit its applicability to WhatsApp and Telegram.

Research [6]: Ahmed et al. (2023)
This research investigates sentiment analysis and machine learning for news classification and fake news detection across digital platforms, achieving 89% accuracy in sentiment classification. It discusses the use of NLP to analyze textual content for deceptive patterns but does not specifically target messaging apps like WhatsApp or Telegram. The study also overlooks behavioral analytics, such as message timing or repetition, and does not address the challenges of lightweight model deployment on mobile devices, which are essential for real-time threat detection in the proposed module.

These studies collectively highlight the potential of NLP and sentiment analysis for detecting suspicious activities but fall short in addressing the specific requirements of WhatsApp and Telegram, such as encrypted data processing, real-time Detection, and mobile-optimized performance. The proposed module aims to bridge these gaps by integrating emotional analytics tailored to messaging platforms, offering a novel approach to fake account detection.

III.2.2-ii Identification of Gaps

The literature on sentiment and emotion analysis for social media and messaging platforms reveals critical shortcomings that the proposed module, focused on analyzing level based emotional tones in WhatsApp and Telegram to detect fake accounts, aims to address. The following gaps are identified:

Real-Time Emotional and Behavioral Analysis in Encrypted Messaging Platforms: Existing research primarily focuses on detecting basic emotions in open social media platforms, neglecting the unique challenges of encrypted messaging apps like WhatsApp and Telegram. These platforms restrict data access and require real-time processing for effective threat detection on mobile devices an area underexplored in current studies. The proposed module addresses this by developing real-time, level-based emotion analysis tailored for encrypted environments.

Integration of Behavioral Features like Message Timing and Frequency: Most studies treat emotions as single, flat categories (e.g., anger, joy, fear) without considering their varying intensity levels. For example, anger can appear as mild irritation, standard anger, or explosive rage. This lack of level-based classification reduces the accuracy of threat detection in messaging contexts, where the severity of emotional intensity often signals manipulation risk. The proposed module fills this gap by incorporating low, medium, and high emotion intensity analysis into chat evaluation.

Mobile-Optimized and Privacy-Preserving Detection Systems: Current research often fails to address the need for lightweight, mobile-optimized models that can perform real-time detection on resource-constrained devices. Additionally, there is limited focus on privacy-preserving techniques for analyzing encrypted messages, particularly for protecting vulnerable users like minors. The proposed module tackles this gap by designing a lightweight, privacy-aware system that ensures efficient performance on mobile devices while adhering to ethical standards

These gaps underscore the need for a novel approach that integrates real-time, level-based emotional analysis, specifically designed for encrypted messaging platforms, to effectively detect fake accounts while prioritizing usability and privacy.

III.2 Feasibility Study

The technical feasibility of the proposed research component involves evaluating the appropriateness of the tools, technologies, and techniques required to develop a machine learning-driven module for analyzing level-based emotional tones in WhatsApp and Telegram messages to detect suspicious or harmful accounts. This module will process chat messages, classify emotions into intensity levels, and issue real-time threat alerts. The feasibility is addressed in the following areas:

III.3.1 Technical Feasibility

III.2.2-i Tools and Technology

- **Python**: This versatile, general-purpose programming language will be used for developing core functionalities of the module, including data preprocessing, text vectorization, and machine learning model implementation. Python's extensive libraries, such as TensorFlow, PyTorch, and Hugging Face Transformers, support natural language processing (NLP) and emotion detection, making it ideal for building and testing level-based emotion classification models.
- **Jupyter Notebook**: This interactive development environment will be employed for prototyping, testing, and visualizing machine learning models. Jupyter Notebook facilitates exploratory data analysis, model fine-tuning, and evaluation of level-based emotional classification, enabling iterative development of the module.
- **Streamlit**: This open-source framework will be utilized to develop interactive dashboards for visualizing detection results. Streamlit allows for the creation of user-friendly interfaces to display real-time emotional insights, such as alerts on high-intensity anger, fear, or sadness, enhancing the module's usability on mobile devices.

III.2.2-ii Algorithm and Architectures

- **BERT (Bidirectional Encoder Representations from Transformers)**: These transformer-based models will be used for level-based emotion classification of messages. Their contextual understanding of text makes them ideal for distinguishing emotions at different intensity levels (e.g., mild anger vs. explosive anger), ensuring accurate risk categorization in WhatsApp and Telegram chats.

- **Random Forest**: This lightweight classification algorithm will be used as a fusion/decision layer to map the outputs of the emotion model into overall threat levels (low, medium, high). Its simplicity and efficiency suit real-time inference on mobile devices.

- **TF-IDF (Term Frequency-Inverse Document Frequency)**: T This text processing technique will be applied in baseline models to extract textual features and support classical ML models for comparison. While deep learning models handle contextual meaning, TF-IDF offers a lightweight alternative for feature extraction.

III.2.2-iii Techniques

- Sentiment and Behavioral Analysis: NLP techniques, leveraging BERT/DeBERTa for emotion classification, will be implemented to detect not only the type of emotion (e.g., anger, fear, sadness, joy) but also its intensity (low, medium, high). This fine-grained approach allows the system to identify emotionally manipulative or harmful messages more precisely.

- Performance Evaluation and Visualization: The module's performance will be assessed using metrics like accuracy, precision, recall, and F1-score to ensure reliable emotion classification. Interactive dashboards, built with Streamlit, will visualize emotion intensity levels and provide intuitive alerts to parents while maintaining privacy compliance.

III.3 PROJECT REQUIREMENTS

The project requirements for the machine learning-driven module to analyze user behavior and emotional tones in WhatsApp and Telegram messages for detecting fake or suspicious accounts are outlined below. These requirements ensure the module effectively processes messages, including converting Singlish to English, performs real-time emotional and behavioral analysis, and delivers user-friendly threat alerts, aligning with the project's goal of enhancing user safety in messaging platforms.

III.3.1 Functional Requirements

- **Message Processing**: The system must retrieve and process messages from WhatsApp and Telegram, ensuring consistency for level-based emotion analysis.

- **Level-Based Emotional Analysis:**The module must analyze messages to extract emotional tones (e.g., sadness, anger, joy, fear, love, surprise) and classify them into intensity levels (low, medium, high). Transformer-based models such as BERT/DeBERTa will be used for fine-grained classification, allowing detection of emotionally

- **Threat Alert Generation and Control:** The system must generate real-time, user-friendly alerts (e.g., visual notifications or warnings) through the mobile application's interface to inform parents of potential risks. Alerts will be tiered (e.g., low, medium, high risk) based on model confidence scores. Additionally, the system must enable parents to block or allow contacts, with blocked contacts being restricted via WhatsApp port/network blocking, ensuring immediate protection against harmful accounts.

III.3.2 Non-Functional Requirements

- **Accuracy**: The module must achieve high accuracy in detecting fake or suspicious accounts, correctly identifying deceptive patterns while minimizing false positives. This ensures reliable differentiation between legitimate and malicious users based on emotional and behavioral analysis.

- **Performance**: The system must provide low-latency, real-time processing of messages to support

seamless operation in dynamic chat environments. Models must be optimized for mobile devices, ensuring efficient performance without excessive computational overhead.

- **Availability**: The module must maintain high availability, operating reliably across diverse network conditions and handling high volumes of messages without degradation in performance, ensuring consistent threat detection for users.

- **Privacy and Security**: The system must adhere to strict privacy standards, processing encrypted messages in a privacy-preserving manner, especially for vulnerable users like minors. It must comply with national cybersecurity guidelines in Sri Lanka, preventing unauthorized access to user data and ensuring ethical handling of sensitive information.

III.4 WORK BREAKDOWN AND GRANT CHART

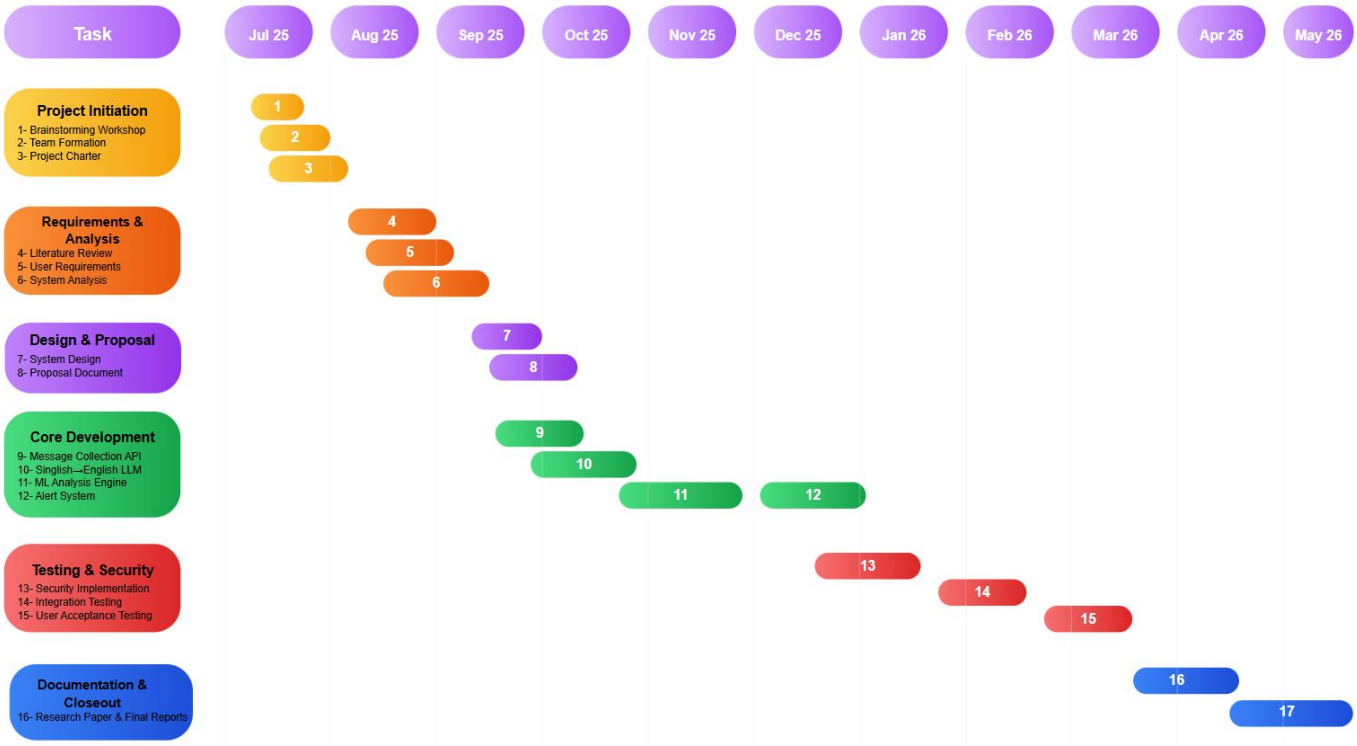

Figure 2 – Grant chart

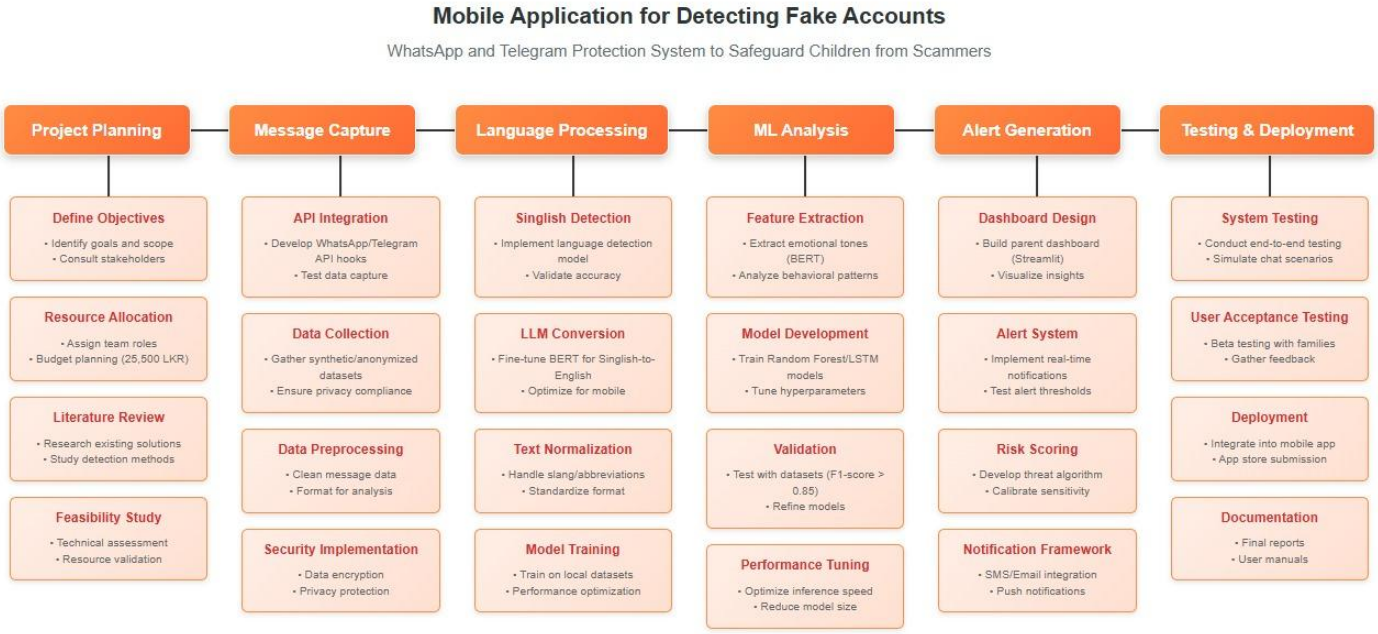

Figure 3 - Work Breakdown Chart

III.5 COMMERCIALIZATION

The proposed machine learning-driven module for level-based emotional analysis in WhatsApp and Telegram messages has strong commercialization potential within the expanding cybersecurity and child online safety market. By integrating advanced features such as fine-grained emotion classification (low, medium, high), media filtering, URL and document threat detection, real-time alerts, and parental control dashboards, the solution offers a powerful tool for enhancing safety on popular messaging platforms. The mobile application is designed for parents, educators, and minors, enabling them to monitor risky interactions, receive instant alerts about potential threats, and manage suspicious contacts through an intuitive interface.

The commercialization strategy is built around accessibility, scalability, and trust. The application will be made available through the Google Play Store and official website download option, ensuring broad reach across platforms. Its lightweight, mobile-optimized design and privacy-preserving architecture will encourage adoption in schools, families, and child safety organizations both in Sri Lanka and globally.

Key Commercialization Points:

- **Target Market**: Educational institutions, child safety organizations, NGOs, cybersecurity providers, and individual users (particularly parents and minors) seeking safe online communication solutions.

- **Unique Selling Proposition:** Unlike generic social media monitoring tools, this application provides level-based emotional analysis (e.g., mild anger vs. explosive anger), real-time detection of fake or harmful accounts, and a parental dashboard with blocking options. This fine-grained, platform-specific approach ensures accurate, regionally relevant protection for vulnerable users.

- **Revenue Model:** The app will follow a tiered pricing structure:

    - Basic ($15): URL & Document Analysis
    - Pro ($30): URL, Document & Chat Analysis
    - Premium ($50): All Features (URL, Document, Chat, Media analysis, and Parent Dashboard)
      This flexible model ensures affordability for individual users while offering advanced features for institutions.

- **Market Entry Strategy**: The first stage of market entry will focus on Sri Lankan schools, universities, and child protection agencies, leveraging partnerships with organizations such as Sri Lanka CERT|CC and NGOs. Demonstrations, school trials, and awareness campaigns will drive adoption. Expansion into global markets will follow, targeting regions with similar WhatsApp and Telegram usage patterns.

- **Promotion Strategy:** Commercialization will be supported by **social media campaigns** (Facebook, Instagram, TikTok), **targeted ads** for parents and schools, and **partnerships with child safety organizations**. Demo versions and school-based trials will be provided to build trust and showcase effectiveness.

IV. BUDGET AND BUDGET JUSTIFICATION

As the proposed module is software-based, no hardware components are required for its implementation. The primary costs are associated with software development, cloud-based computing resources for model training and testing, and publication expenses. The budget also accounts for miscellaneous expenses such as internet connectivity and stationery for documentation. The following table outlines the estimated costs for developing and implementing the module.

Table IV-1 - Budget Analysis

Type	Cost
Internet and Cloud Services	6000 LKR
Publication Cost	10000 LKR
Stationary	4500 LKR
Software & Licenses	15000 LKR
Total	35000 LKR

Budget Justification:

- **Internet and Cloud Services (6,000 LKR):** Covers subscription fees for cloud platforms (e.g., Google Colab, AWS, or Hugging Face) used for training and testing level-based emotion classification models. This ensures access to sufficient computational resources for processing datasets and optimizing models to run efficiently on mobile devices.
- **Publication Cost (10,000 LKR):** Accounts for expenses related to submitting the research to academic journals, conferences, or open-access repositories to disseminate findings. Publishing results helps validate the research academically and supports future commercialization opportunities.
- **Stationery (4,500 LKR):** Includes costs for printing, documentation, and other materials needed for project planning, stakeholder consultations, and preparing the final research report.
- **Software & Licenses (15,000 LKR):** Covers costs for required software tools, paid APIs, and licenses for libraries used in model development and integration, including platforms that support real-time NLP and mobile deployment.

Summary

This research proposes a mobile application to protect children from scammers on WhatsApp and Telegram by detecting fake or harmful accounts. It captures chat messages and applies level-based emotional analysis using machine learning (ML) techniques such as BERT and Logistic Regression to classify emotions into intensity levels (low, medium, high). Based on the detected emotional risks, the system issues real-time alerts via a parent dashboard, with options to block suspicious contacts by restricting WhatsApp communication. The methodology includes message capture, ML-based emotion classification, threat level decision making, and alert generation, using Python and optimized models for mobile deployment. Addressing a gap in WhatsApp/Telegram safety research, particularly the lack of level-based emotional analysis for child protection the project demonstrates technical feasibility with tools like Hugging Face Transformers, TensorFlow Lite, and Streamlit dashboards. The commercialization strategy targets schools, parents, and child safety organizations through a freemium pricing model, supported by a 35,000 LKR budget for cloud services, software, and publication, aligning with Sri Lanka's cybersecurity goals.

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